

A Rough Guide

for the project upon Energy Stock Price Forecasting

— *being the private notes and standing questions of* —

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7 Python scripts 17 technical indicators 3 models trained 3 working weeks ~1,250 trading days

I. Three weeks at a glance

Wk	Mon	Tue	Wed	Thu	Fri	Sat	Sun
1	Setup Data	EDA Feat	Base LSTM	Back test	Debug Review	Code clean	Git push
2	README draft	Report update	Tune LSTM	More stocks	Study LSTM	Sharpe derive	Prep Q&A
3	Oil feat	GRU compare	Docs clean	Resume update	Mock interview	Final push	—

II. Week by week

Week 1 — Foundation

Day 1: Set up project folder, venv, install dependencies. Download 5yr NTPC data via yfinance.
Day 2: Full EDA (6 plots, ADF test). Feature engineering (17 indicators). Baselines (LR + RF). Start LSTM.
Day 3-4: Debug LSTM target scaling. Backtesting — long-only strategy, Sharpe, max drawdown. GitHub push.
Day 5-7: Code review, add docstrings. Understand why LR beats LSTM. Write up key learnings.

Week 2 — Polish & study

Day 8-9: Write README.md with results and plots. Update project report PDF.
Day 10-11: Hyperparameter tuning (seq_len, hidden_dim). Add more stocks (Reliance, ONGC).
Day 12-14: Study LSTM gates and cell state. Derive Sharpe ratio. Prepare interview answers.

Week 3 — Extend & prep

Day 15-17: Optional — crude oil as external feature. GRU comparison.
Day 18-19: Final cleanup, pin repo on GitHub, update resume.
Day 20-21: Mock interview — explain project in 5 min. Practice key Q&A.;

III. Open questions & decisions

Things to resolve as the project progresses.

- Q1.** Is predicting next-day close the right target, or should I predict returns / direction instead?
- Q2.** Should I use NTPC alone, or include 2-3 more energy stocks for a multi-stock model?
- Q3.** Is 5 years of daily data sufficient, or should I also try intraday (hourly) data?
- Q4.** For backtesting: simple long-only strategy, or something more sophisticated (long/short)?
- Q5.** Should the report include derivations (Sharpe ratio, LSTM backprop) or keep it applied?
- Q6.** Any additional baseline model worth adding beyond Linear Regression and Random Forest?
- Q7.** Is a Transformer comparison within scope, or save it for a follow-up project?
- Q8.** How to present the LR-beats-LSTM finding — as a limitation or as an insight?

IV. Reading list & resources

Priority	Resource	Status
Core	Hochreiter & Schmidhuber 1997 (original LSTM paper)	To read
Core	Fischer & Krauss 2018 (LSTM for financial markets)	To read
Core	Murphy 1999 — Technical Analysis of Financial Markets (Ch. 1-4)	To read
Applied	Brownlee 2018 — Deep Learning for Time Series	To read
Applied	Selvin et al. 2017 — LSTM vs CNN for stock prediction	To read
Applied	Nelson et al. 2017 — LSTM for stock movement	To read
Ref	PyTorch LSTM documentation	Used
Ref	ta library documentation	Used
Ref	scikit-learn user guide	Used

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